Madeleine Vercher

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Education

Cornell University, Duffield College of Engineering, Ithaca, NY
M.Eng. in Engineering Management — Expected December 2027
B.S. in Mechanical Engineering — Expected May 2027
Dyson Business Minor for Engineers | Cumulative GPA: 3.19

Relevant Coursework: Heat Transfer, Mechatronics, Fluid Mechanics, Statics, Mechanical Design, System Dynamics, Mechanics of Materials, Organizational Behavior, Business Simulation

Skills: Creo, Fusion, Autodesk Inventor, Arduino, C, Python, MATLAB, Excel, SAP, Windchill, 3D Printing, Laser Cutting, A3 Problem Solving, Time Studies | Spanish (Advanced)

Engineering Projects:

Competitive Robotics Design Project, *Cornell University*

- Built an autonomous cube-collection robot on a three-person team using Arduino, C, sensors, motors, and laser-cut components
- Designed the circuitry, CAD base, guide walls, and detachable mechanism; iterated the design for reliable deployment and earned selection for an ASML exhibition match

Satellite Attitude Control System Analysis, *Cornell University*

- Modeled a simplified satellite attitude control system by deriving rotational dynamics and designing a feedback controller while evaluating disturbances such as solar radiation pressure and drag

Professional Experience

Manufacturing and Quality Engineering Intern, *Jabil,* Elmira, NY — May 2026 - August 2026

- Led cross-functional projects in process improvement, waste reduction, tooling design, and workflow standardization using Creo, Excel, SAP, and Windchill
- Designed and prototyped manufacturing aids, analyzed production data, and updated drawings, process sheets, and engineering change documentation
- Conducted time studies and layout analysis for an A3 grinder-cell project, identifying improvements projected to reduce operator travel by 35% and qualifying to begin the Lean Bronze Certification process

Undergraduate Teaching Assistant, *Cornell University,* Ithaca, NY — January 2026 - May 2026

- Lead lab sections for Mechanical Design, supporting students with CAD modeling, prototyping, fabrication, and design methodology.

Lead Teaching Assistant, *Cornell University*, Ithaca, NY — January 2025 - Present

- Serve as the sole Teaching Assistant for the college-level class, leading discussions and office hours while grading coursework and supporting students in energy and environmental systems

Autodesk Design and Make Ambassador, *Autodesk*, Remote — September 2024 - Present

- Organize campus workshops and outreach events introducing students to CAD, Autodesk design software, and additive manufacturing while promoting engineering design technologies through digital content

Leadership Experience

Vice President of Learning and Development, *Kappa Delta*, Ithaca, NY — May 2025 - May 2026

- Design and manage the new member education program for a 175-member chapter, overseeing a 54-person new member class from bid day through initiation while ensuring compliance with Cornell Panhellenic and national organization requirements

President, *Big Red Bears*, Ithaca, NY — August 2023 - Present

- Lead Cornell's mascot program, managing 40 volunteers and coordinating 100+ campus and community events per semester, while overseeing safety training, scheduling logistics, and official branding decisions